

Ionatan Kuperwajs

Curriculum Vitae

ikuperwajs@princeton.edu | ionatankuperwajs.github.io | Princeton University

EMPLOYMENT

Princeton University , Princeton, NJ Postdoctoral Research Associate Department of Computer Science Advisor: Thomas L. Griffiths	2024-Present
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EDUCATION

New York University , New York, NY Ph.D. in Neural Science Thesis: Cognitive mechanisms of complex planning Advisor: Wei Ji Ma	2018-2024
Macalester College , St. Paul, MN B.A. in Neuroscience, Computer Science, and Mathematics Honors in Mathematics, Magna Cum Laude Advisor: Andrew Beveridge	2014-2018

AWARDS AND FELLOWSHIPS

NYU Graduate School of Arts and Sciences Dean's Outstanding Dissertation Award	2024
National Science Foundation Graduate Research Fellowship	2020-2023
Henry Mitchell McCracken Fellowship	2018
Phi Beta Kappa National Honor Society Member	2018
Macalester College Neuroscience Department Outstanding Graduate Award	2018
Macalester College Dean's List	2014-2018
Macalester College DeWitt Wallace Distinguished Scholar	2014

PUBLICATIONS

Preprints

KM Collins, CE Zhang, G Todd, L Ying, MB da Costa, R Liu, P Sharma, A Weller, **I Kuperwajs**, L Wong, JB Tenenbaum, and TL Griffiths (2025). Evaluating language models' evaluations of games. *arXiv*. [📄](#)

I Kuperwajs, B van Opheusden, EM Russek, and TL Griffiths (2024). Learning from rewards and social information in naturalistic strategic behavior. *PsyArXiv*. [📄](#)

I Kuperwajs, MK Ho, and WJ Ma (2024). Heuristics for meta-planning from a normative model of information search. *PsyArXiv*. [📄](#)

Journal articles

I Kuperwajs, EM Russek, MG Mattar, WJ Ma, and TL Griffiths (2025). Looking deeper into the algorithms underlying human planning. *Trends in Cognitive Sciences*. [📄](#)

I Kuperwajs, HH Schütt, and WJ Ma (2023). Using deep neural networks as a guide for modeling human planning. *Scientific Reports*. [📄](#)

B van Opheusden, **I Kuperwajs**, G Galbiati, Z Bnaya, Y Li, and WJ Ma (2023). Expertise increases planning depth in human gameplay. *Nature*. 

Conference proceedings

I Kuperwajs, EM Russek, L Schut, Y Sagiv, MG Mattar, WJ Ma, and TL Griffiths (2025). Exploring resource-rational planning under time pressure in online chess. *Proceedings of the 47th Annual Meeting of the Cognitive Science Society*. 

L Wong, T Mills, **I Kuperwajs**, KM Collins, and TL Griffiths (2025). Meta-reasoning: deciding which game to play, which problem to solve, and when to quit. *Proceedings of the 47th Annual Meeting of the Cognitive Science Society*. 

I Kuperwajs, HH Schütt, and WJ Ma (2022). Improving a model of human planning via large-scale data and deep neural networks. *Proceedings of the 44th Annual Meeting of the Cognitive Science Society*. 

I Kuperwajs and WJ Ma (2022). A joint analysis of dropout and learning functions in human decision-making with massive online data. *Proceedings of the 44th Annual Meeting of the Cognitive Science Society*. 

I Kuperwajs and WJ Ma (2021). Planning to plan: a Bayesian model for optimizing the depth of decision tree search. *Proceedings of the 43rd Annual Meeting of the Cognitive Science Society*. 

I Kuperwajs, B van Opheusden, and WJ Ma (2019). Prospective planning and retrospective learning in a large-scale combinatorial game. *Cognitive Computational Neuroscience*. 

SUMMER SCHOOLS AND INTERNSHIPS

Simons Computational Neuroscience Imbizo, Cape Town, South Africa	2018
Howard Hughes Medical Institute Janelia Undergraduate Scholars Program, Ashburn, VA	2017
New York University Center for Neural Science NSF REU, New York, NY	2016

INVITED TALKS

Reinforcement Learning and Decision Making (Workshop), Trinity College Dublin	2025
Parallel Distributed Processing Seminar, Princeton University	2024
Computational Cognitive Neuroscience Lab, UC Berkeley	2023
Computational Cognitive Science Lab, Princeton University	2023
Cognitive Science Society, Toronto, Canada (2 talks)	2022
Cognitive Science Society, University of Vienna	2021
Center for Neural Science Seminar, New York University	2020
Concepts and Categories Seminar, New York University	2019
Artificial and Biological Computation Lab, New York University	2019
Sensorimotor Learning Group, Columbia University	2019

CONFERENCE POSTERS

Neural Information Processing Systems (Workshop), San Diego, United States	2025
Cognitive Science Society, San Francisco, United States (2 posters)	2025
Association for the Advancement of Artificial Intelligence (Bridge), Vancouver, Canada	2024
Cognitive Computational Neuroscience, Oxford University	2023
Minds, Brains, and Machines, New York University	2023
Reinforcement Learning and Decision Making, Brown University (3 posters)	2022
Workshop on Scaling Cognitive Science, Princeton University	2019
Cognitive Computational Neuroscience, Technical University of Berlin	2019

TEACHING

New York University, Teaching Assistant

NEURL-GA 2201: Mathematical Tools for Neural and Cognitive Science F 19

Macalester College, Teaching Assistant

COMP 221: Algorithm Design and Analysis F 17, S 18

PSYC 180: Brain, Mind, and Behavior F 16

COMP 123: Core Concepts in Computer Science S 16, F 16

MENTORSHIP

Graduate students

Ido Ben-Artzi, Tel Aviv University (Neuroscience) 2025-Present

Hanbo Xie, Georgia Institute of Technology (Psychology) 2025-Present

Lisa Schut, University of Oxford (Computer Science) 2024-Present

Undergraduate students

Kirill Sirik, Princeton University (Operations Research and Financial Engineering) 2025-Present

Issac Li, Princeton University (Computer Science) 2025-Present

Brandon Lee, Princeton University (Computer Science) 2025

SERVICE

Ad hoc reviewing

Cognitive Computational Neuroscience

Cognitive Science

Cognitive Science Society

Communications Psychology

Nature Communications

Nature Human Behaviour

Philosophical Transactions of the Royal Society A

Scientific Reports

Workshop organization

Meta-reasoning: deciding which game to play, which problem to solve, and when to quit [↗](#) 2025

Co-organizer, Cognitive Science Society, San Francisco, United States

Chess as a bridge between human cognition and artificial intelligence [↗](#) 2025

Co-organizer, Reinforcement Learning and Decision Making, Trinity College Dublin

ADVOCACY AND OUTREACH

President, Scientist Action and Advocacy Network (ScAAN) [↗](#) 2018-2023

Workshop on the Climate Crisis, Cognitive Computational Neuroscience, Oxford University 2023

Workshop on Environmental Justice, Ocean Sciences Meeting 2022

Workshop on Evidence-Based Advocacy, American Geophysical Union, New Orleans, LA 2021

Workshop on Science Activism, Rockefeller University 2021

Panel on Science Activism, Growing Up in Science 2020

REFERENCES

Thomas L. Griffiths
Professor, Princeton University
Departments of Psychology and Computer Science
`tomg@princeton.edu`

Wei Ji Ma
Professor, New York University
Center for Neural Science and Department of Psychology
`weijima@nyu.edu`

Heiko H. Schütt
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Department of Behavioural and Cognitive Sciences
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Mark K. Ho
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